Sampling Distributions & Central Limit Theorem

Section 1: The Central Limit Theorem (CLT)

1 What does the Central Limit Theorem state about the sampling distribution of the sample mean $\bar{x}$?

a. When sample size n is sufficiently large ($n \geq 30$), the sampling distribution of $\bar{x}$ is approximately normal, regardless of the population shape

b. As n increases, the population becomes normally distributed

c. The sample standard deviation approaches the population standard deviation

d. Every random sample will have an identical mean

2 If a population is already normally distributed with mean μ and standard deviation σ, what is the shape of the sampling distribution of $\bar{x}$ for a small sample of $n = 5$?

a. Exactly normal

b. Uniform

c. Skewed

d. Cannot be determined without $n \geq 30$

3 Does the Central Limit Theorem apply to the shape of the individual sample data values?

a. No, it applies only to the distribution of sample statistics across repeated sampling, not the sample data itself

b. Yes, individual observations become normal as n grows

c. Only if the data has no outliers

d. Yes, this is known as the Law of Large Numbers

Section 2: Mean and Standard Error of the Sample Mean

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4 A population has mean $\mu = 80$ and standard deviation $\sigma = 20$. For random samples of size $n = 25$, what are the mean and standard error of $\bar{x}$?

a. $\mu_{\bar{x}} = 80, \sigma_{\bar{x}} = 4$

b. $\mu_{\bar{x}} = 80, \sigma_{\bar{x}} = 20$

c. $\mu_{\bar{x}} = 16, \sigma_{\bar{x}} = 4$

d. $\mu_{\bar{x}} = 80, \sigma_{\bar{x}} = 0.8$

5 To cut the standard error of the sample mean $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ in half, the sample size must be:

a. Quadrupled ($4n$)

b. Doubled ($2n$)

c. Halved

d. Increased by 10

6 When using the formula $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$, what condition must be satisfied if sampling without replacement?

a. The 10% condition: sample size $n \leq 0.10N$ (population size)

b. The Large Counts Condition: $np \geq 10$

c. $n \geq 30$

d. The population must be symmetric

Section 3: Sampling Distribution of Sample Proportion p-hat

7 In a large population, 40% of adults support a referendum ($p = 0.40$). For random samples of $n = 100$, what are the mean and standard deviation of $\hat{p}$?

a. $\mu_{\hat{p}} = 0.40, \sigma_{\hat{p}} = 0.049$

b. $\mu_{\hat{p}} = 40, \sigma_{\hat{p}} = 4.9$

c. $\mu_{\hat{p}} = 0.40, \sigma_{\hat{p}} = 0.24$

d. $\mu_{\hat{p}} = 0.60, \sigma_{\hat{p}} = 0.049$

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8 Which conditions justify modeling the sampling distribution of $\hat{p}$ with a normal distribution?

a. $np \geq 10$ and $n(1-p) \geq 10$

b. $n \geq 30$

c. $n \leq 0.10N$

d. $p = 0.50$

9 Why is the sample proportion $\hat{p}$ called an unbiased estimator of p?

a. The mean of its sampling distribution equals the true population parameter p

b. It never makes a prediction error

c. Its standard deviation is zero

d. It is computed from a simple random sample

Section 4: Calculations with Sampling Distributions

10 A tire manufacturer knows tire lifespan is normal with $\mu = 50000$ miles and $\sigma = 4000$ miles. What is the probability that an SRS of 16 tires has a mean lifespan under 48000 miles?

a. $P(Z < -2.0) \approx 0.0228$

b. $P(Z < -0.5) \approx 0.3085$

c. $P(Z > 2.0) \approx 0.9772$

d. 0.0500

11 How does increasing sample size affect the bias and variability of a sample statistic?

a. Variability decreases, while bias remains unchanged

b. Both bias and variability decrease

c. Bias decreases, variability remains unchanged

d. Both remain unchanged

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12 If X_1 and X_2 are independent sample means from two populations, what is the standard error of $(\bar{x}_1 - \bar{x}_2)$?

a. $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

b. $\frac{\sigma_1}{\sqrt{n_1}} - \frac{\sigma_2}{\sqrt{n_2}}$

c. $\sqrt{\frac{\sigma_1}{n_1} + \frac{\sigma_2}{n_2}}$

d. $\frac{\sigma_1 + \sigma_2}{\sqrt{n_1 + n_2}}$

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